

Sharp spectral constants for scaled q -numerical ranges: the quadratic case and an exact completion reduction

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Abstract

For $0 < |q| \leq 1$, put

$$\Omega_q(A) = q^{-1}W_q(A), \quad C_q = \max \left\{ 1, \frac{2|q|}{1 + \sqrt{1 - |q|^2}} \right\}.$$

O'Loughlin and Rani conjectured that $\|p(A)\| \leq C_q \max_{\Omega_q(A)} |p|$ for every square matrix A of order at least two and every polynomial p . We prove the conjecture, with the sharp constant, for every 2×2 matrix and, more generally, for every finite-dimensional matrix whose minimal polynomial has degree at most two. The result also holds for finite orthogonal direct sums of such matrices. For matrices of every size we prove the sharp norm equivalence

$$w_q(A) \geq \min \left\{ |q|, \frac{1 + \sqrt{1 - |q|^2}}{2} \right\} \|A\|,$$

which settles the conjecture for all affine polynomials. The proof combines the exact q -ellipse with a two-point Schur-class extremal calculation and a quadratic-operator canonical form. For the unresolved general case we derive the precise parameterized positive-real completion inequality that an extension of the sharp numerical-range argument must satisfy. This separates the remaining work into two explicit statements: an operator completion theorem and an algebra-sensitive boundary-measure construction. The manuscript deliberately labels those two statements as open; no claim of a proof of the general conjecture is made.

1 Statement, normalization, and sharpness

Throughout, $n \geq 2$. For a matrix $A \in M_n(\mathbb{C})$ and $q \in \mathbb{C}$, $0 < |q| \leq 1$, define

$$W_q(A) = \{ \langle Ax, y \rangle : \|x\| = \|y\| = 1, \langle x, y \rangle = q \}, \quad \Omega_q(A) = \frac{1}{q} W_q(A).$$

Changing the phase of y shows that $\Omega_q(A) = \Omega_{|q|}(A)$. We therefore write

$$r = |q|, \quad \tau = \sqrt{1 - r^2}, \quad \kappa = \frac{1 + \tau}{r}, \quad \rho = \frac{2}{\kappa} = \frac{2r}{1 + \tau}.$$

The proposed constant is

$$C_r = \max\{1, \rho\}.$$

Notice that $\rho = 1$ exactly when $r = 4/5$.

Problem 1.1 (Scaled q -numerical-range conjecture). Prove that for every $A \in M_n(\mathbb{C})$, $n \geq 2$, and every complex polynomial p ,

$$\|p(A)\| \leq C_r \sup_{z \in \Omega_r(A)} |p(z)|.$$

Both terms in C_r are necessary. The constant polynomial $p \equiv 1$ forces a constant at least one. For

$$J = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$$

the exact q -ellipse formula gives

$$\Omega_r(J) = \frac{\kappa}{2} \overline{\mathbb{D}}.$$

Taking $p(z) = z$ gives the ratio $2/\kappa = \rho$.

We record the support-function formula that explains the geometry. Writing $y = rx + \tau z$, where $z \perp x$ and $\|z\| = 1$, gives

$$h_{\Omega_r(A)}(e^{i\theta}) = \sup_{\|x\|=1} \left[\operatorname{Re} \langle e^{-i\theta} Ax, x \rangle + \frac{\tau}{r} \left\| P_{x^\perp}(e^{-i\theta} Ax) \right\| \right]. \quad (1)$$

Indeed, after x is fixed, the phase of z can be chosen to make the second summand real and nonnegative. The q -numerical range is convex [5]; hence comparison of support functions in (1) gives $W(A) \subseteq \Omega_r(A)$.

Proposition 1.2 (The normal case). *Problem 1.1 holds with constant 1 whenever A is normal.*

Proof. The spectrum of A is contained in $W(A)$, hence in $\Omega_r(A)$. The spectral theorem therefore gives

$$\|p(A)\| = \max_{\lambda \in \sigma(A)} |p(\lambda)| \leq \sup_{z \in \Omega_r(A)} |p(z)|.$$

□

2 The conjecture for two-by-two matrices

For $0 < r \leq 1$, set

$$\tau = \sqrt{1 - r^2}, \quad \kappa = \kappa(r) := \frac{1 + \tau}{r},$$

and

$$C_r := \max \left\{ 1, \frac{2}{\kappa} \right\} = \max \left\{ 1, \frac{2r}{1 + \sqrt{1 - r^2}} \right\}.$$

Thus $\kappa \geq 1$, with $\kappa = 2$ precisely when $r = 4/5$; moreover, $C_r = 1$ precisely for $0 < r \leq 4/5$. We write

$$\Omega_r(A) := \frac{1}{r} W_r(A).$$

The standard phase covariance of the q -numerical range gives $\Omega_q(A) = \Omega_{|q|}(A)$, so it is enough to consider real $q = r$.

Theorem 2.1. *Let $A \in M_2(\mathbb{C})$, let $0 < |q| \leq 1$, and let p be a complex polynomial. Then*

$$\|p(A)\| \leq \max \left\{ 1, \frac{2|q|}{1 + \sqrt{1 - |q|^2}} \right\} \sup_{z \in \Omega_q(A)} |p(z)|.$$

For each fixed q , this constant is best possible uniformly over $A \in M_2(\mathbb{C})$.

We first isolate the disk functional-calculus calculation that will be used in the distinct-eigenvalue case.

Lemma 2.2 (A two-point Schur-class calculation). *For $\sigma \geq 0$, let*

$$B_\sigma = \begin{pmatrix} 1 & 2\sigma \\ 0 & -1 \end{pmatrix}, \quad R = R_\sigma := \sqrt{1 + \sigma^2} + \sigma.$$

Then $\|B_\sigma\| = R$, and for every $0 < a < 1$,

$$\sup_{\substack{g \in H^\infty(\mathbb{D}) \\ \|g\|_\infty \leq 1}} \|g(aB_\sigma)\| = \max\{1, aR\}.$$

Proof. The positive matrix $B_\sigma^* B_\sigma$ has determinant 1 and trace $2 + 4\sigma^2 = R^2 + R^{-2}$. Its eigenvalues are therefore R^2 and R^{-2} , which proves $\|B_\sigma\| = R$.

The lower bound follows by taking $g \equiv 1$ and $g(z) = z$. Put $T = aB_\sigma$. If $aR \leq 1$, then T is a contraction, and von Neumann's inequality gives the upper bound.

Suppose now that

$$C := aR \geq 1.$$

Since $B_\sigma^2 = I$, if $u = g(a)$ and $v = g(-a)$, then

$$g(T) = \begin{pmatrix} u & \sigma(u - v) \\ 0 & v \end{pmatrix}. \quad (2)$$

The set

$$\mathcal{K}_a := \{(g(a), g(-a)) : g \in H^\infty(\mathbb{D}), \|g\|_\infty \leq 1\}$$

is compact and convex. By the two-point Pick theorem, its nonconstant boundary is described by equality in

$$\left| \frac{u - v}{1 - \bar{u}v} \right| \leq \frac{2a}{1 + a^2}.$$

Equality in Schwarz–Pick implies that the corresponding pair is obtained from a disk automorphism. If $|u| = 1$ or $|v| = 1$, the pair instead comes from a unimodular constant. Consequently every extreme point of $\mathcal{K}_a$ is obtained from either a unimodular constant or a disk automorphism. Since the norm in (2) is a convex function of (u, v) , it is enough to check these functions.

Unimodular constants are immediate. Up to an irrelevant unimodular factor, a disk automorphism has the form

$$\psi_\zeta(z) = \frac{z - \zeta}{1 - \bar{\zeta}z}, \quad |\zeta| < 1.$$

Let

$$A_\zeta = T - \zeta I, \quad D_\zeta = I - \bar{\zeta}T,$$

and consider the Hermitian matrix

$$\Delta := C^2 D_\zeta^* D_\zeta - A_\zeta^* A_\zeta.$$

Write $z = |\zeta|^2$ and $x = \operatorname{Re} \zeta$. Direct calculation gives

$$\operatorname{tr} \Delta = z(C^4 + a^4 - 2) + C^2 - \frac{a^4}{C^2}, \quad (3)$$

$$\begin{aligned} \det \Delta = (C^2 - 1)[2(C^2 - a^2)(1 + a^2)z - 4a^2(C^2 - 1)x^2 \\ - (1 - a^4)(C^2 + 1)z^2]. \end{aligned} \quad (4)$$

If $C^4 + a^4 - 2 \geq 0$, then (3) is nonnegative by setting $z = 0$. If $C^4 + a^4 - 2 < 0$, then $0 \leq z < 1$ gives

$$\operatorname{tr} \Delta \geq C^4 + a^4 - 2 + C^2 - \frac{a^4}{C^2} = \frac{(C^2 - 1)(C^4 + 2C^2 + a^4)}{C^2} \geq 0.$$

Since $x^2 \leq z$ and $z^2 \leq z$, the expression in square brackets in (4) is bounded below by

$$z[2(C^2 - a^2)(1 + a^2) - 4a^2(C^2 - 1) - (1 - a^4)(C^2 + 1)] = z(C^2 - 1)(1 - a^2)^2.$$

Thus $\operatorname{tr} \Delta \geq 0$ and $\det \Delta \geq 0$, so $\Delta \succeq 0$. It follows that

$$\|\psi_\zeta(T)\| = \|A_\zeta D_\zeta^{-1}\| \leq C.$$

This proves the required upper bound. $\square$

Remark 2.3 (Algebra audit). The identities (3) and (4), as well as both displayed factorizations used to prove their nonnegativity, were independently checked in exact symbolic algebra. They were also spot-checked numerically. No algebraic identity in Lemma 2.2 remains unchecked.

We shall also need the coalescing-eigenvalue version.

Lemma 2.4 (The square-zero limit). *Let*

$$J = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}.$$

For every $c \geq 0$,

$$\sup_{\substack{g \in H^\infty(\mathbb{D}) \\ \|g\|_\infty \leq 1}} \|g(cJ)\| = \max\{1, c\}.$$

Proof. Again, $g \equiv 1$ and $g(z) = z$ give the lower bound. If $c \leq 1$, the upper bound is von Neumann's inequality. Suppose $c \geq 1$. Since $J^2 = 0$,

$$g(cJ) = \begin{pmatrix} g(0) & c g'(0) \\ 0 & g(0) \end{pmatrix}.$$

Put $b = |g(0)|$. Schwarz–Pick gives $|g'(0)| \leq 1 - b^2$. A 2×2 upper triangular matrix $\begin{pmatrix} b & d \\ 0 & b \end{pmatrix}$ with $b \leq 1 \leq c$ has norm at most c if and only if

$$(c^2 - b^2)^2 \geq c^2 |d|^2.$$

Here $|d| \leq c(1 - b^2)$, and

$$c^2 - b^2 - c^2(1 - b^2) = b^2(c^2 - 1) \geq 0.$$

Hence $\|g(cJ)\| \leq c$. $\square$

Proof of Theorem 2.1. The affine covariance

$$\Omega_r(\alpha A + \beta I) = \alpha \Omega_r(A) + \beta \quad (\alpha, \beta \in \mathbb{C})$$

shows that the assertion is invariant under affine changes of the matrix.

Assume first that A has two distinct eigenvalues. Schur triangularization, followed by an affine change and a diagonal unitary similarity, reduces A to

$$B_\sigma = \begin{pmatrix} 1 & 2\sigma \\ 0 & -1 \end{pmatrix}, \quad \sigma \geq 0.$$

Put

$$v = \sqrt{1 + \sigma^2}, \quad R = v + \sigma = \|B_\sigma\|.$$

The Hermitian and skew-Hermitian parts of B_σ are

$$H = \begin{pmatrix} 1 & \sigma \\ \sigma & -1 \end{pmatrix}, \quad K = \begin{pmatrix} 0 & -i\sigma \\ i\sigma & 0 \end{pmatrix}.$$

Thus

$$\operatorname{tr}(H^2) = 2(1 + \sigma^2), \quad \operatorname{tr}(K^2) = 2\sigma^2, \quad \operatorname{tr}(HK) = 0.$$

The two-by-two q -elliptical range formula [1, Lemma 6] therefore shows that $\Omega_r(B_\sigma)$ is the ellipse centered at the origin with semiaxes

$$A_r = \frac{v + \tau\sigma}{r}, \quad B_r = \frac{\sigma + \tau v}{r}. \quad (5)$$

These axes satisfy

$$A_r^2 - B_r^2 = 1, \quad A_r + B_r = \kappa(v + \sigma) = \kappa R.$$

Consequently $\Omega_r(B_\sigma) = E_\eta$, where

$$\eta := \kappa R$$

and E_η is the confocal ellipse

$$E_\eta = \left\{ x + iy : \frac{x^2}{((\eta + \eta^{-1})/2)^2} + \frac{y^2}{((\eta - \eta^{-1})/2)^2} \leq 1 \right\}.$$

Suppose first that $\eta > 1$. Let

$$\phi_\eta : E_\eta^\circ \longrightarrow \mathbb{D}$$

be normalized by $\phi_\eta(0) = 0$ and $\phi'_\eta(0) > 0$. Symmetry and uniqueness imply that ϕ_η is odd. Write

$$a_\eta := \phi_\eta(1) \in (0, 1).$$

We claim that

$$a_\eta \leq \frac{2}{\eta}. \quad (6)$$

For completeness, let $\xi = \log \eta$. The classical Schwarz map in the normalization of Kanas–Sugawa [3, Equation (2.1) and the inverse formula following it] chooses $m \in (0, 1)$ so that

$$\frac{\pi}{2} \frac{K(m')}{K(m)} = 2\xi$$

where we use the modulus convention

$$K(m) = \int_0^{\pi/2} \frac{dt}{\sqrt{1 - m^2 \sin^2 t}}, \quad m' = \sqrt{1 - m^2},$$

and gives

$$\phi_\eta(w) = \sqrt{m} \operatorname{sn} \left(\frac{2K(m)}{\pi} \arcsin w, m \right).$$

In particular, $a_\eta = \sqrt{m}$. If Q denotes the elliptic nome, then

$$Q = \exp \left(-\pi \frac{K(m')}{K(m)} \right) = e^{-4\xi} = \eta^{-4}$$

and the standard nome identities [2, Equations 22.2.1–22.2.2] give

$$m = \frac{\vartheta_2(0, Q)^2}{\vartheta_3(0, Q)^2}, \quad a_\eta = \frac{\vartheta_2(0, Q)}{\vartheta_3(0, Q)}.$$

Using the theta series [2, Equations 20.2.2–20.2.3],

$$\vartheta_2(0, Q) = 2Q^{1/4} \sum_{n=0}^{\infty} Q^{n(n+1)}, \quad \vartheta_3(0, Q) = 1 + 2 \sum_{n=1}^{\infty} Q^{n^2}.$$

Since $0 < Q < 1$ and $n(n+1) \geq n^2$,

$$\sum_{n=0}^{\infty} Q^{n(n+1)} \leq \sum_{n=0}^{\infty} Q^{n^2} < \vartheta_3(0, Q).$$

As $Q^{1/4} = \eta^{-1}$, this proves (6).

Because $B_\sigma^2 = I$ and ϕ_η is odd,

$$\phi_\eta(B_\sigma) = a_\eta B_\sigma.$$

For a polynomial p , define

$$g = p \circ \phi_\eta^{-1}.$$

The conformal map extends continuously to the boundary of the ellipse, so $g \in H^\infty(\mathbb{D})$ and $\|g\|_{\infty, \mathbb{D}} = \|p\|_{\infty, E_\eta}$. The composition rule for the holomorphic functional calculus and Lemma 2.2 now give

$$\begin{aligned} \|p(B_\sigma)\| &= \|g(a_\eta B_\sigma)\| \\ &\leq \max\{1, a_\eta R\} \|p\|_{\infty, E_\eta} \\ &\leq \max\left\{1, \frac{2R}{\eta}\right\} \|p\|_{\infty, E_\eta} \\ &= \max\left\{1, \frac{2}{\kappa}\right\} \|p\|_{\infty, \Omega_r(B_\sigma)}. \end{aligned}$$

The only distinct-eigenvalue case with $\eta = 1$ is $r = 1$ and $\sigma = 0$. Then B_σ is normal and $\Omega_1(B_\sigma) = [-1, 1]$, so the spectral theorem gives the stronger constant 1.

Suppose next that A has a repeated eigenvalue. If A is scalar, the assertion is immediate. Otherwise, affine covariance reduces the problem to

$$J = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}.$$

The same two-by-two q -elliptical range formula gives

$$\Omega_r(J) = \left\{ z \in \mathbb{C} : |z| \leq \frac{\kappa}{2} \right\}.$$

Put $c = 2/\kappa$ and, for a polynomial p , set $g(z) = p((\kappa/2)z)$. Then

$$p(J) = g(cJ), \quad \|g\|_{\infty, \mathbb{D}} = \|p\|_{\infty, \Omega_r(J)}.$$

Lemma 2.4 yields

$$\|p(J)\| \leq \max \left\{ 1, \frac{2}{\kappa} \right\} \|p\|_{\infty, \Omega_r(J)}.$$

This completes the proof of the inequality.

Finally, for $A = J$, the constant polynomial $p \equiv 1$ forces every spectral constant to be at least 1, while $p(z) = z$ gives the ratio

$$\frac{\|J\|}{\sup_{z \in \Omega_r(J)} |z|} = \frac{2}{\kappa}.$$

Hence the best uniform constant over $M_2(\mathbb{C})$ is exactly C_r . □

3 Quadratic matrices in arbitrary finite dimension

The two-by-two result extends without loss to every finite-dimensional matrix whose minimal polynomial has degree at most two. In fact, both the scaled q -numerical range and every polynomial norm are already determined by one extremal 2×2 block.

Proposition 3.1 (Exact reduction to an extremal block). *Let $A \in M_n(\mathbb{C})$ be nonscalar and suppose that*

$$(A - aI)(A - bI) = 0$$

for some $a, b \in \mathbb{C}$, where $a = b$ is allowed. Then there is a number $d \geq 0$ such that, for

$$A_0 = \begin{pmatrix} a & d \\ 0 & b \end{pmatrix},$$

the following identities hold for every $0 < |q| \leq 1$ and every complex polynomial f :

$$\Omega_q(A) = \Omega_q(A_0), \quad \|f(A)\| = \|f(A_0)\|.$$

Proof. The canonical form for a quadratic operator [4, Theorem 1.1] gives an orthogonal decomposition

$$\mathbb{C}^n = \mathcal{H}_1 \oplus \mathcal{H}_1 \oplus \mathcal{H}_2$$

with respect to which

$$A \simeq \begin{pmatrix} aI_{\mathcal{H}_1} & P \\ 0 & bI_{\mathcal{H}_1} \end{pmatrix} \oplus \gamma I_{\mathcal{H}_2}. \tag{7}$$

Here P is positive semidefinite, $\gamma \in \{a, b\}$, and $\mathcal{H}_1 \neq \{0\}$. When $a = b$, P has zero kernel. Set

$$d := \|P\|.$$

Since the space is finite dimensional, P has a unit eigenvector for the eigenvalue d . The quadratic-operator elliptical range theorem [4, Theorem 3.1 and Corollary 3.3] therefore gives

$$W_r(A) = W_r(A_0) \quad (0 \leq r \leq 1).$$

After division by r , and then use of the standard phase covariance $\Omega_q = \Omega_{|q|}$, this proves $\Omega_q(A) = \Omega_q(A_0)$ for $0 < |q| \leq 1$.

It remains to prove the norm identity. Let

$$d = d_1 \geq d_2 \geq \cdots \geq d_m \geq 0$$

be the eigenvalues of P , counted with multiplicity, where $m = \dim \mathcal{H}_1$. Diagonalizing P in both copies of $\mathcal{H}_1$ in (7) shows that

$$A \simeq A_{d_1} \oplus \cdots \oplus A_{d_m} \oplus \gamma I_{\mathcal{H}_2}, \quad A_t := \begin{pmatrix} a & t \\ 0 & b \end{pmatrix}. \quad (8)$$

For a polynomial f , put

$$\delta_f := \begin{cases} \frac{f(a) - f(b)}{a - b}, & a \neq b, \\ f'(a), & a = b. \end{cases}$$

The polynomial functional calculus gives, in both cases,

$$f(A_t) = \begin{pmatrix} f(a) & t \delta_f \\ 0 & f(b) \end{pmatrix}. \quad (9)$$

For fixed $u, v, w \in \mathbb{C}$, define

$$M_t = \begin{pmatrix} u & tw \\ 0 & v \end{pmatrix}, \quad t \geq 0.$$

If

$$S_t := |u|^2 + |v|^2 + t^2 |w|^2,$$

then the trace and determinant of $M_t^* M_t$ are S_t and $|uv|^2$, respectively. Hence

$$\|M_t\|^2 = \frac{S_t + \sqrt{S_t^2 - 4|uv|^2}}{2}. \quad (10)$$

The right-hand side is nondecreasing in S_t , and S_t is nondecreasing in t . Thus

$$0 \leq t_1 \leq t_2 \implies \|M_{t_1}\| \leq \|M_{t_2}\|.$$

Applying this observation to (9) yields

$$\|f(A_{d_j})\| \leq \|f(A_d)\| = \|f(A_0)\| \quad (1 \leq j \leq m).$$

Moreover, since $\gamma \in \{a, b\}$,

$$|f(\gamma)| \leq r(f(A_0)) \leq \|f(A_0)\|,$$

where $r(\cdot)$ denotes the spectral radius. Taking norms in (8) now gives

$$\|f(A)\| = \max \left\{ \max_{1 \leq j \leq m} \|f(A_{d_j})\|, |f(\gamma)| \right\} = \|f(A_0)\|,$$

as required. $\square$

Theorem 3.2 (The conjecture for quadratic matrices). *Let $A \in M_n(\mathbb{C})$, $n \geq 2$, have minimal polynomial of degree at most two. Then, for every $0 < |q| \leq 1$ and every complex polynomial p ,*

$$\|p(A)\| \leq \max \left\{ 1, \frac{2|q|}{1 + \sqrt{1 - |q|^2}} \right\} \sup_{z \in \Omega_q(A)} |p(z)|.$$

For every $n \geq 2$ and every fixed q , this constant is best possible uniformly over the quadratic matrices in $M_n(\mathbb{C})$.

Proof. If the minimal polynomial has degree one, then A is scalar and

$$\|p(A)\| = |p(A)| = \sup_{z \in \Omega_q(A)} |p(z)|.$$

Suppose that the minimal polynomial has degree two. It either has two distinct roots, in which case A is annihilated by $(z - a)(z - b)$ with $a \neq b$, or it has a repeated root, in which case $(A - aI)^2 = 0$. Both cases are covered by Proposition 3.1. Thus there is a 2×2 matrix A_0 such that

$$\Omega_q(A) = \Omega_q(A_0), \quad \|p(A)\| = \|p(A_0)\|.$$

The asserted inequality follows directly from Theorem 2.1.

This argument also covers all degeneracies of the extremal block. If the two roots are distinct and $d = 0$, then A_0 is normal. If the roots coincide, then $A_0 - aI$ is square-zero; the nonscalar assumption forces $d > 0$. Scalar matrices were handled separately.

To see sharpness in every dimension $n \geq 2$, take

$$A = J \oplus 0_{n-2}, \quad J = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}.$$

This matrix is quadratic, and Proposition 3.1 reduces it to J itself. As in the proof of Theorem 2.1, the constant polynomial gives the lower bound 1, while $p(z) = z$ gives the lower bound

$$\frac{2|q|}{1 + \sqrt{1 - |q|^2}}.$$

□

4 Reducing direct sums

The preceding result is stable under finite orthogonal direct sums. This gives a class considerably larger than the matrices with quadratic minimal polynomial.

Theorem 4.1 (Direct sums of quadratic matrices). *Suppose*

$$A = A_1 \oplus \cdots \oplus A_m$$

and every A_j has minimal polynomial of degree at most two. Then Problem 1.1 holds for A , with the sharp uniform constant C_r .

Proof. If the j -th summand has dimension at least two, vectors supported in that summand show that

$$W_r(A_j) \subseteq W_r(A), \quad \Omega_r(A_j) \subseteq \Omega_r(A).$$

Theorem 3.2 then gives

$$\|p(A_j)\| \leq C_r \sup_{\Omega_r(A_j)} |p| \leq C_r \sup_{\Omega_r(A)} |p|.$$

If $A_j = [\lambda]$ is one dimensional, then $\lambda \in W(A) \subseteq \Omega_r(A)$, and therefore

$$\|p(A_j)\| = |p(\lambda)| \leq \sup_{\Omega_r(A)} |p|.$$

Taking the maximum over j , and using $\|p(A)\| = \max_j \|p(A_j)\|$, proves the assertion. Sharpness already occurs for a summand equal to J . $\square$

5 The linear-polynomial checkpoint

Testing Problem 1.1 with $p(z) = z$ suggests the inequality

$$w_r(A) \geq \min \left\{ r, \frac{1 + \sqrt{1 - r^2}}{2} \right\} \|A\|, \quad w_r(A) := \sup_{z \in W_r(A)} |z|. \quad (11)$$

The constants are forced respectively by scalar matrices and by rank-one nilpotents with orthogonal initial and final vectors.

We first prove the scalar inequality underlying the result. A phase and a unitary similarity put an arbitrary 2×2 matrix in the Pauli form

$$A = (u + iv)I + h\sigma_z + ik\sigma_x, \quad h \geq k \geq 0.$$

To see this, subtract half the trace and write the traceless part as $H + iK$, with H, K traceless Hermitian. A scalar phase rotates the real two-plane spanned by H, K ; choose it so that the rotated pair is Hilbert–Schmidt orthogonal and order the pair by norm. The Pauli-vector identification of the traceless Hermitian matrices with $\mathbb{R}^3$, together with the $SO(3)$ action induced by unitary similarity, then sends the pair to $h\sigma_z, k\sigma_x$. Writing $X = |u|$, $Y = |v|$, one has

$$\|A\|^2 = X^2 + Y^2 + h^2 + k^2 + 2\sqrt{h^2X^2 + k^2Y^2 + h^2k^2}, \quad (12)$$

$$w_r(A)^2 \geq \max \left\{ r^2Y^2 + (rX + h + \tau k)^2, r^2X^2 + (rY + k + \tau h)^2 \right\}. \quad (13)$$

The first identity follows by computing the larger eigenvalue of A^*A . The second follows from the exact two-by-two q -ellipse formula [1, Lemma 6]. The two quantities in (13) are the squared moduli of the horizontal and vertical extreme points of the exact q -ellipse. Hence the sharp linear checkpoint is implied by the elementary scalar inequality

$$\begin{aligned} & \max \left\{ r^2Y^2 + (rX + h + \tau k)^2, r^2X^2 + (rY + k + \tau h)^2 \right\} \\ & \geq \min \left\{ r^2, \frac{(1 + \tau)^2}{4} \right\} \left(X^2 + Y^2 + h^2 + k^2 + 2\sqrt{h^2X^2 + k^2Y^2 + h^2k^2} \right). \end{aligned} \quad (14)$$

This reduction is useful independently of the nonlinear functional calculus: any proposed proof of the full conjecture must, at a minimum, account for the phase transition already visible in (14).

Lemma 5.1 (The sharp scalar inequality). *Let $0 < r \leq 1$, put*

$$s = (1 - r^2)^{1/2}, \quad c = \min \left\{ r, \frac{1+s}{2} \right\},$$

and suppose that $X, Y \geq 0$ and $h \geq k \geq 0$. Define

$$\begin{aligned} N^2 &= X^2 + Y^2 + h^2 + k^2 + 2\sqrt{h^2X^2 + k^2Y^2 + h^2k^2}, \\ M_1^2 &= r^2Y^2 + (rX + h + sk)^2, \\ M_2^2 &= r^2X^2 + (rY + k + sh)^2. \end{aligned}$$

Then

$$\max\{M_1, M_2\} \geq cN.$$

The constant is sharp: if $c = r$, equality is attained by $h = k = 0$ and $(X, Y) \neq (0, 0)$; if $c = (1 + s)/2$, equality is attained by $X = Y = 0$ and $h = k > 0$.

Proof. Set

$$\alpha = h + sk, \quad \beta = k + sh.$$

We first show that there is a number $\lambda \in [0, 1]$ such that

$$\lambda\alpha \geq ch, \quad (1 - \lambda)\beta \geq ck. \tag{15}$$

If $h = k = 0$, this is immediate. Suppose $h > 0$, and write $t = k/h \in [0, 1]$. When a quotient below has both numerator and denominator equal to zero, it is to be read as zero. We have

$$\frac{h}{\alpha} + \frac{k}{\beta} = \frac{1}{1 + st} + \frac{t}{t + s}.$$

For $s > 0$, the derivative of the right-hand side is

$$s \left(\frac{1}{(t + s)^2} - \frac{1}{(1 + st)^2} \right) \geq 0,$$

because

$$t + s \leq 1 + st \iff (1 - s)(1 - t) \geq 0.$$

Consequently,

$$\frac{h}{\alpha} + \frac{k}{\beta} \leq \frac{2}{1 + s}.$$

The same estimate holds when $s = 0$: the left-hand side is 2 for $t > 0$, and is 1 under the above convention when $t = 0$. Since $c \leq (1 + s)/2$, it follows that

$$\frac{ch}{\alpha} + \frac{ck}{\beta} \leq 1.$$

Thus the interval

$$\left[\frac{ch}{\alpha}, 1 - \frac{ck}{\beta} \right]$$

is nonempty, and any λ in this interval satisfies (15).

Using

$$\begin{aligned} M_1^2 &= r^2(X^2 + Y^2) + \alpha^2 + 2rX\alpha, \\ M_2^2 &= r^2(X^2 + Y^2) + \beta^2 + 2rY\beta, \end{aligned}$$

we obtain

$$\begin{aligned} \max\{M_1^2, M_2^2\} &\geq \lambda M_1^2 + (1 - \lambda)M_2^2 \\ &= r^2(X^2 + Y^2) + \lambda\alpha^2 + (1 - \lambda)\beta^2 \\ &\quad + 2r(\lambda X\alpha + (1 - \lambda)Y\beta). \end{aligned}$$

Now $c \leq r$, and (15) gives

$$r\lambda\alpha \geq rch \geq c^2h, \quad r(1 - \lambda)\beta \geq rck \geq c^2k.$$

Moreover, by convexity of the square and (15),

$$\lambda\alpha^2 + (1 - \lambda)\beta^2 \geq (\lambda\alpha + (1 - \lambda)\beta)^2 \geq c^2(h + k)^2.$$

It follows that

$$\max\{M_1^2, M_2^2\} \geq c^2(X^2 + Y^2 + (h + k)^2 + 2hX + 2kY). \quad (16)$$

Finally, all three quantities hX, kY, hk are nonnegative, so

$$\sqrt{h^2X^2 + k^2Y^2 + h^2k^2} \leq hX + kY + hk.$$

Therefore

$$N^2 \leq X^2 + Y^2 + h^2 + k^2 + 2hX + 2kY + 2hk = X^2 + Y^2 + (h + k)^2 + 2hX + 2kY.$$

Together with (16), this proves $\max\{M_1, M_2\} \geq cN$.

For sharpness, if $h = k = 0$, then

$$N = (X^2 + Y^2)^{1/2}, \quad M_1 = M_2 = r(X^2 + Y^2)^{1/2}.$$

Thus equality holds whenever $c = r$. On the other hand, if $X = Y = 0$ and $h = k > 0$, then

$$N = 2h, \quad M_1 = M_2 = (1 + s)h,$$

so equality holds whenever $c = (1 + s)/2$. □

Theorem 5.2 (Sharp norm equivalence for the q -numerical radius). *For every $A \in M_n(\mathbb{C})$, $n \geq 2$, and every $0 < r \leq 1$,*

$$w_r(A) \geq \min \left\{ r, \frac{1 + \sqrt{1 - r^2}}{2} \right\} \|A\|.$$

The constant is best possible in every dimension at least two.

Proof. We first prove the assertion for $B \in M_2(\mathbb{C})$. Multiplication by a scalar phase and unitary similarity put B in the Pauli form used above. The exact norm formula is (12). The q -ellipse of B is centered at $r(u + iv)$, has a horizontal semiaxis $h + \tau k$, and a vertical semiaxis $k + \tau h$. Its horizontal and vertical extreme points give (13). Lemma 5.1, with

$$c = \min \left\{ r, \frac{1 + \tau}{2} \right\},$$

therefore yields $w_r(B) \geq c\|B\|$.

For a general $A \in M_n(\mathbb{C})$, $n \geq 2$, choose a unit right singular vector x such that $\|Ax\| = \|A\|$, and put $\mathcal{E} = \text{span}\{x, Ax\}$. If $\dim \mathcal{E} = 1$, enlarge $\mathcal{E}$ to any two-dimensional subspace. The compression

$$B = P_{\mathcal{E}} A|_{\mathcal{E}}$$

satisfies

$$\|B\| \geq \|Bx\| = \|Ax\| = \|A\|,$$

while the reverse inequality is automatic; hence $\|B\| = \|A\|$. Every admissible pair of vectors in $\mathcal{E}$ is also admissible in the full space, so $W_r(B) \subseteq W_r(A)$. Consequently

$$w_r(A) \geq w_r(B) \geq c\|B\| = c\|A\|.$$

Scalar matrices show sharpness of the branch $c = r$. For the other branch the q -ellipse calculation gives $w_r(J) = (1 + \tau)/2$. In higher dimensions take $A = J \oplus 0$. Proposition 3.1 gives $W_r(A) = W_r(J)$, so these are sharp examples in every dimension at least two. $\square$

Since

$$\sup_{z \in \Omega_r(A)} |z| = \frac{w_r(A)}{r},$$

Theorem 5.2 proves Problem 1.1 for $p(z) = z$, with the sharp constant $C_r = r/c$. More generally, for $p(z) = \alpha z + \beta$, affine covariance gives

$$\Omega_r(p(A)) = p(\Omega_r(A)).$$

Applying Theorem 5.2 to $p(A)$ therefore proves the conjecture for every affine polynomial, again with C_r . This is only a checkpoint for the nonlinear functional calculus: q -numerical ranges do not in general obey a polynomial spectral-mapping inclusion.

6 A parameterized positive-real completion

The proof that the numerical range is a 2-spectral set uses a positive-real completion of the resolvent. The correct interpolation of that completion for the present problem is

$$F_\rho(w, T) = \left(1 - \frac{2}{\rho}\right) I + \frac{2}{\rho} (I - wT)^{-1}, \quad 0 < \rho \leq 2.$$

Without an adjoint-algebra correction, the condition $\text{Re } F_\rho(w, T) \succeq 0$ is the standard Herglotz characterization of a ρ -contraction.

Proposition 6.1 (A necessary Gramian inequality). *Let $T \in M_n(\mathbb{C})$ have distinct eigenvalues in $\overline{\mathbb{D}}$. Suppose $H : \mathbb{D} \rightarrow M_n(\mathbb{C})$ is analytic and*

$$H(0) = I, \quad \operatorname{Re} H(w) \succeq 0, \quad H(w) - F_\rho(w, T) \in \operatorname{alg}(T^*).$$

For $0 < t < 1$, define

$$P_t = \sum_{m=0}^{\infty} t^{2m} T^{*m} T^m, \quad Q_t = \sum_{m=0}^{\infty} t^m T^{*m} T^m, \quad Y_t = Q_t - P_t.$$

Then

$$\frac{2}{1-t} Y_t - Y_t P_t - P_t Y_t - (2-\rho)(P_t^2 - P_t) \succeq 0. \quad (17)$$

Proof. Diagonalize $T = S\Lambda S^{-1}$, put $G = S^* S$, and work first in the eigenvector coordinates. Polynomial interpolation identifies $\operatorname{alg}(T^*)$ with the diagonal algebra in these coordinates. Thus the transformed completion has the form

$$\tilde{H}(w) = (1-k)G + kG(I - w\Lambda)^{-1} + \Theta(w)G, \quad k = \frac{2}{\rho},$$

where $\Theta(w)$ is diagonal analytic and $\Theta(0) = 0$.

The matrix Herglotz theorem implies that

$$\mathcal{K}_H(w, z) = \frac{H(w) + H(z)^*}{1 - w\bar{z}}$$

is a positive kernel. Use its positivity at

$$w_i = t\bar{\lambda}_i \quad (1 \leq i \leq n), \quad w_0 = 0.$$

In eigenvector coordinates let

$$P_{ij} = \frac{G_{ij}}{1 - t^2 \bar{\lambda}_i \lambda_j}, \quad Q_{ij} = \frac{G_{ij}}{1 - t \bar{\lambda}_i \lambda_j}, \quad Y = Q - P.$$

For an arbitrary $u \in \mathbb{C}^n$, use the coordinate vectors $u_i e_i$ at w_i and the compensating vector

$$v = -G^{-1}Pu$$

at the origin. Exactly as in the unparameterized completion argument, this choice cancels every value of the unknown diagonal function Θ . Explicitly, the entire correction contribution to the kernel quadratic form is

$$2 \operatorname{Re} \sum_{i=1}^n \bar{u}_i \theta_i(w_i) ((Gv)_i + (Pu)_i) = 0.$$

The remaining main block, cross block, and origin block are respectively

$$2P + \frac{2k}{1-t}Y, \quad (2-k)G + kQ, \quad 2G.$$

Here one uses the scalar partial-fraction identity

$$\frac{1}{(1-ta)(1-t^2a)} = \frac{1}{1-t} \left(\frac{1}{1-ta} - \frac{t}{1-t^2a} \right).$$

Substitution of $v = -G^{-1}Pu$ into the positive-kernel quadratic form, followed by division by k , gives

$$\frac{2}{1-t}Y - YG^{-1}P - PG^{-1}Y + (2-\rho)(P - PG^{-1}P) \succeq 0.$$

After congruence by $G^{-1/2}$, put

$$\tilde{T} = G^{1/2}\Lambda G^{-1/2}.$$

The three balanced matrices are precisely the series in the statement with T replaced by $\tilde{T}$. The polar decomposition $S = UG^{1/2}$ gives $T = U\tilde{T}U^*$, so conjugating back by U yields (17) as stated. $\square$

Corollary 6.2 (The square-zero model of the Gramian inequality). *Let $T^2 = 0$, and suppose that (17) holds for every $0 < t < 1$. Then*

$$\|T\| \leq \begin{cases} \rho, & 1 \leq \rho \leq 2, \\ \sqrt{\rho/(2-\rho)}, & 0 < \rho \leq 1. \end{cases}$$

In particular, $\|T\| \leq \max\{1, \rho\}$.

Proof. Put $X = T^*T$. Then

$$P_t = I + t^2X, \quad Y_t = t(1-t)X.$$

Equation (17) reduces to

$$\rho X - t(2-\rho t)X^2 \succeq 0.$$

Consequently

$$\|T\|^2 = \|X\| \leq \inf_{0 < t < 1} \frac{\rho}{t(2-\rho t)}.$$

If $\rho \geq 1$, the denominator is maximized at $t = 1/\rho$, giving $\|T\| \leq \rho$. If $\rho \leq 1$, its supremum is $2-\rho$, attained as $t \uparrow 1$, and the second bound follows. $\square$

Problem 6.3 (Parameterized completion theorem). Under the hypotheses of Proposition 6.1, prove

$$\|T\| \leq \max\{1, \rho\}.$$

Corollary 6.2 shows that the Gramian inequality has at least the required strength on the square-zero model, and is exactly sharp there when $\rho \geq 1$. (The simple-spectrum hypothesis in Proposition 6.1 is a device for deriving the inequality, not an assumption in the corollary.) For a general power sequence, (17) alone has not yet yielded the asserted norm bound; a proof must exploit either the full positive kernel or the coupling of (17) over all t .

7 The exact geometric completion still required

Let B have simple spectrum and let f be holomorphic on a neighbourhood of $\Omega_r(B)$, normalized so that $\|f\|_{\Omega_r(B)} \leq 1$, with $\text{alg}(f(B)) = \text{alg}(B)$, and put $T = f(B)$. The ordinary double-layer calculus produces a positive-real completion with parameter 2. To obtain the conjectured constant one needs the following strengthened, algebra-aware statement.

Problem 7.1 (Geometric q -completion). With

$$\rho = \frac{2r}{1 + \sqrt{1 - r^2}},$$

construct an analytic $H : \mathbb{D} \rightarrow M_n(\mathbb{C})$ such that

$$H(0) = I, \quad \operatorname{Re} H(w) \succeq 0, \quad H(w) - F_\rho(w, T) \in \operatorname{alg}(T^*).$$

A sufficient, stronger formulation is to construct a positive operator-valued boundary measure M on $\partial\Omega_r(B)$ having mass ρI and satisfying, for every holomorphic h ,

$$\int h dM - h(B) \in \operatorname{alg}(B^*).$$

Indeed, in that case

$$H(w) = \frac{1}{\rho} \int_{\partial\Omega_r(B)} \frac{1 + wf(z)}{1 - wf(z)} dM(z)$$

has positive real part and $H(0) = I$. Expanding the Cayley kernel in powers of w , and using $\operatorname{alg}(B^*) = \operatorname{alg}(f(B)^*)$, gives

$$H(w) - F_\rho(w, f(B)) \in \operatorname{alg}(f(B)^*).$$

Proposition 7.2 (Exact obstruction to scalar stripping). *Let $0 < r < 1$ and $A = \operatorname{diag}(-1, 1)$. If one subtracts the pointwise smallest eigenvalue from the standard double-layer measure on $\partial\Omega_r(A)$, the remaining mass is*

$$\rho_{\text{strip}}(r)I, \quad \rho_{\text{strip}}(r) = 2 - \frac{4}{\pi} \arccos r.$$

This is strictly larger than the desired parameter

$$\rho_r = \frac{2r}{1 + \sqrt{1 - r^2}}.$$

Thus pointwise scalar stripping cannot produce the measure required in Problem 7.1, even for this normal 2×2 matrix.

Proof. Put $s = \sqrt{1 - r^2}$. The scaled q -range is the ellipse parametrized by

$$\zeta(\phi) = \frac{\cos \phi + is \sin \phi}{r}, \quad 0 \leq \phi \leq 2\pi.$$

Direct substitution in the scalar double-layer formula shows that the two diagonal densities, with respect to $d\phi$, are

$$m_\lambda(\phi) = \frac{1}{\pi} \operatorname{Re} \frac{-i\zeta'(\phi)}{\zeta(\phi) - \lambda}, \quad m_+(\phi) = \frac{s}{\pi(1 - r \cos \phi)}, \quad m_-(\phi) = \frac{s}{\pi(1 + r \cos \phi)}.$$

Each has integral 2. The scalar mass that can be removed while preserving pointwise positivity is therefore

$$\begin{aligned} a(r) &= \int_0^{2\pi} \min\{m_+(\phi), m_-(\phi)\} d\phi \\ &= \frac{s}{\pi} \int_0^{2\pi} \frac{d\phi}{1 + r|\cos \phi|} = \frac{4}{\pi} \arccos r. \end{aligned}$$

The remaining mass is $2 - a(r)$, proving the first formula.

Write $r = \sin \theta$, $0 < \theta < \pi/2$. Then

$$\rho_{\text{strip}}(r) = \frac{4\theta}{\pi}, \quad \rho_r = 2 \tan \frac{\theta}{2}.$$

The strict convexity of $\tan x$ on $[0, \pi/4]$ places its graph strictly below the chord joining $(0, 0)$ and $(\pi/4, 1)$. Consequently

$$\tan \frac{\theta}{2} < \frac{2\theta}{\pi},$$

which proves $\rho_{\text{strip}}(r) > \rho_r$. □

The ordinary double-layer measure has mass $2I$. Thus the issue is not merely to preserve positivity, but to lower the mass to ρI while retaining all of the stated holomorphic moments modulo $\text{alg}(B^*)$. Proposition 7.2 shows that scalar subtraction does not suffice; the adjoint-algebra freedom is the additional structure available for a successful construction.

Proposition 7.3 (Conditional closure of the conjecture). *If Problems 6.3 and 7.1 hold, then Problem 1.1 holds for every finite-dimensional matrix.*

Proof. For simple spectrum and an algebra-generating auxiliary f , the geometric completion supplies the hypotheses of the parameterized completion theorem, hence

$$\|f(B)\| \leq \max\{1, \rho\} = C_r.$$

Let p be an arbitrary polynomial. If $\lambda_1, \dots, \lambda_n$ are the distinct eigenvalues of B , then

$$p_\varepsilon(z) = p(z) + \varepsilon z$$

takes distinct values on the λ_i 's for all but finitely many ε . Hence $\text{alg}(p_\varepsilon(B)) = \text{alg}(B)$. Apply the preceding estimate to

$$f_\varepsilon = \frac{p_\varepsilon}{\|p_\varepsilon\|_{\Omega_r(B)}}$$

and let $\varepsilon \rightarrow 0$. This proves the conjectured estimate for every polynomial when B has simple spectrum.

Finally choose matrices $B_j \rightarrow A$ with simple spectra. The same compact parameter space of admissible pairs in the definitions gives

$$d_H(W_r(B_j), W_r(A)) \leq \|B_j - A\|, \quad d_H(\Omega_r(B_j), \Omega_r(A)) \leq \frac{\|B_j - A\|}{r}.$$

Therefore $p(B_j) \rightarrow p(A)$, while

$$\sup_{\Omega_r(B_j)} |p| \longrightarrow \sup_{\Omega_r(A)} |p|.$$

Passing to the limit proves the result for A . □

8 Conclusion

The sharp constant and its phase transition are now proved for all quadratic matrices and their reducing direct sums. The two-dimensional proof is not merely a check of a special example: the exact extremizers I and J already force both branches of the conjectured constant. In the general case the argument has been reduced to Problems 6.3 and 7.1. The first is an operator-theoretic completion question; the second is a geometric moment problem on the boundary of the scaled q -numerical range. Closing only one of them is insufficient, and the scalar-floor version of the second loses essential algebraic information.

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